Digital Electronics Design
(25ECSF101)

TW & SL Activity Report
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Name	Kartik S Shivshetty
Roll No.	632
SRN	01FE25BCS387
Semester	I
Division	F

1. Problem Statement

To design and simulate a **Car Speed Monitoring System** that represents vehicle speed through clock pulses using sequential logic, and compares the measured value with predefined speed limits.

The system should automatically detect over-speed conditions and disable further counting using logic-based clock control, thereby providing a clear indication of speed violations.

2. Hardware and Software requirements

Hardware (Simulated): clock, counters, comparators, logic gates, seven-segment display
Software: Logisim

3. Brief description of tool used

a. *Why the simulation tool was chosen?*

It was chosen for its simplicity, real-time simulation, and availability of built-in components required for timing-based systems.

b. *Features, limitations, and scope?*

Features: Real-time simulation, modular design
Limitations: Ideal timing, no real sensor noise
Scope: Can be extended for traffic monitoring and violation detection

c. *Application in the Microproject?*

The tool is used to simulate how a vehicle's speed is calculated digitally using counters and clocks, demonstrating practical application of **timing, counters, and comparators** taught in Digital Electronics.

d. *Use of tool commands, libraries, or modules relevant to the project?*

1.Clock module for time measurement

2.Counter module for counting clock pulses

3.Comparator for speed limit checking

4.Seven-segment decoder and display

e. Skills gained?

1. Digital circuit design

2.Understanding of counters and timing circuits

3.Logical problem decomposition

4.Simulation and debugging skills

4. Working description

The clock signal acts as a reference source and is used to generate uniform pulses that correspond to vehicle speed. These pulses are applied to a clock-gated counter, where the number of pulses counted represents the vehicle's speed in digital form. The counter output is simultaneously fed to a multi-digit display unit and to comparison logic.

Two comparators continuously monitor the counter output. The first comparator checks whether the measured speed exceeds the permissible speed limit, while the second comparator checks for a higher maximum threshold indicating a critical over-speed condition. The outputs of both comparators are combined using an OR gate.

When either comparator detects a limit violation, the OR gate output disables the clock signal before it reaches the counter. This halts further counting, freezes the displayed speed value, and activates an alert indication. Thus, the system ensures immediate detection and response to over-speed conditions using purely digital logic.

5. Block diagram

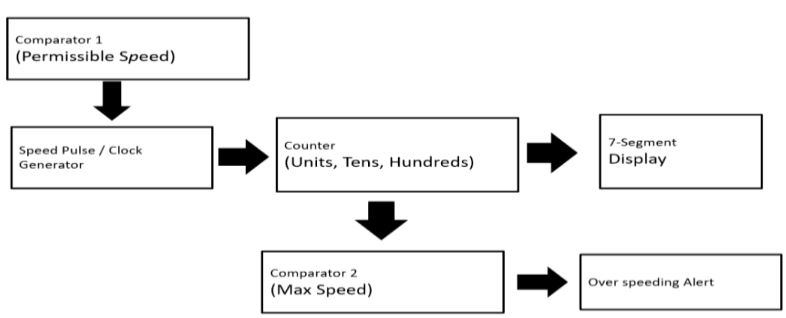

Fig 1.: Block Diagram

6. Circuit Diagram

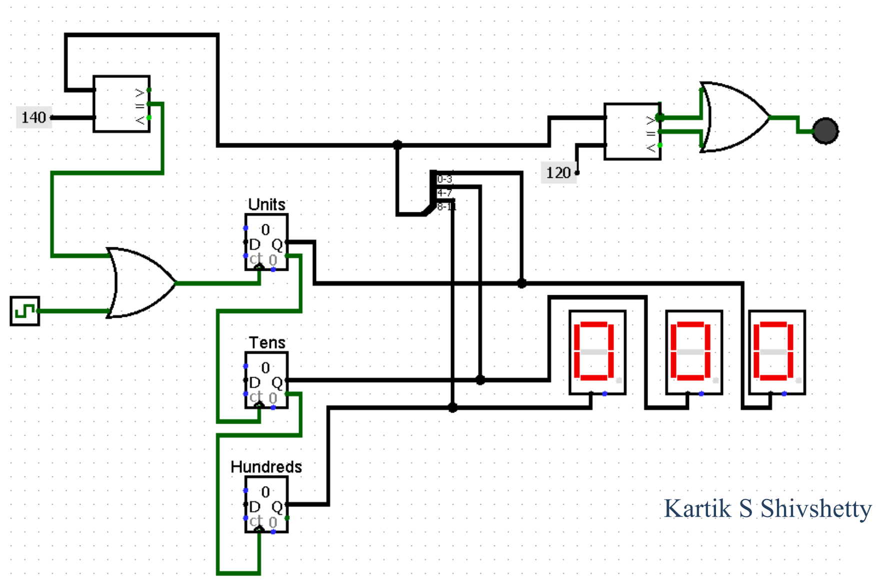

Fig 2.:Circuit Diagram

7. Correlation to theory

This project directly applies fundamental concepts of Digital Electronics. The clock generator represents the use of **timing signals** to control sequential circuits. The counter section demonstrates the practical application of **synchronous counters**, where clock pulses are used to represent a physical quantity in digital form.

The use of **comparators** relates to magnitude comparison theory, where digital values are evaluated against predefined reference levels. The **OR gate** implements combinational logic to combine multiple decision outputs into a single control signal. Clock gating illustrates the concept of **control logic**, where the operation of a sequential circuit is enabled or disabled based on logical conditions.

Overall, the system integrates **sequential logic, combinational logic, and control mechanisms**, reinforcing theoretical concepts taught in counters, comparators, logic gates, and clock-controlled digital systems.

8. Conclusion

The car speed monitoring system was successfully designed and simulated using digital logic components. The system represents vehicle speed by counting clock pulses and continuously compares the measured value with predefined speed limits. The use of clock gating ensures an immediate response to over-speed conditions by halting further counting and activating an alert indication.

This project effectively demonstrates the integration of **sequential circuits, combinational logic, and control mechanisms** in solving a real-world problem. It also highlights how digital electronics concepts such as counters, comparators, and logic gates can be applied to monitoring and safety applications. The simulation provides a clear understanding of system behavior and forms a strong foundation for future hardware implementation.